(3) The automatic water valve on the non-lifting injector may be binding on the valve spindle so that the tank head pressure cannot open the valve.

(4) Injector tubes may be worn or tubes of improper size applied when repairs are made.

(5) As the tank water temperature increases, the injector capacity decreases.

If the injector does not take up all the water at the overflow, and it is found necessary to cut down on the water valve, it may be due to the following causes:

(1) Worn injector tubes.

(2) Lack of forcing pressure due to reduced boiler pressure or obstruction in forcing the steam tube.

(3) Resistance in the branch pipe due to scale deposits or obstruction in the boiler or intermediate check valves.

When an injector spills water and it is necessary to cut down on the steam valve in order to make the injector take up, the cause is lack of water supply. In this case, look for the trouble in the feed supply line from the tank to the injector.

If the injector spills water and steam at the overflow, the cause is a worn steam nozzle. In this case, there is more steam than the water will take care of, because the water has reached a higher temperature. Therefore, the increase in steam supply, due to the worn steam tube, causes the water to expand in the combining chamber and to spill out through the overflow. This may be especially true of the lifting type of injector in the summertime.

Never mix tube numbers, sizes or capacities in an injector body, nor allow tube sizes to vary from standards and limits of wear. Most roads prefer to use tubes made by the manufacturer of the injector. Injector tubes require a high degree of precision and necessitate more than merely reaming an orifice or throat or applying a taper from throat to mouth of an injector tube. The proper materials must be used, the design of the tube must adhere to correct thermodynamics, and the workmanship to a precision standard.

Repairs to Ohio Lifting Injectors. Refer to Fig. 26 and pay particular attention to the following: The steam valve *13* and its seat against the valve seat of the steam tube *16*. The steam valve washer *14* between steam valve *13* and steam valve spindle *11;* the primer *15* and its seat against the valve seat of steam tube *16;* the steam nozzle *16* flange and its seat against the tube seat of the body *1* (back part) of the injector. It is essential that there is a steam tight joint between the body and flange of steam nozzle *16*. If this joint leaks, it will result in two things: Heating of the feed water at the injector connection, causing trouble in priming and the possibility of eventually resulting in a steam cut through the flange on the body and threads, destroying both the steam tube and the body.

Keep the original clearance and alignment between all tubes in the injector body.

Repairs to Chicago Non-Lifting Injectors. Here again, it is important to maintain in proper condition the steam nozzles *A* and *C,* Fig. 23. If these nozzles are cut, they permit the entrance of more steam than the hot water can take care of. In other words, there is not enough volume of water to take care of the excess steam from a worn steam nozzle or forcing tube. This used excess steam wastes power, water, and fuel.

Keep the lime off the tube seats and maintain *A* and *C* with good seat in the injector body. Keep these two tubes securely fastened together so that they will not come loose.

The lifting tube *D,* the rear combining tube *E,* the forward combining tube *F,* and the delivery tube *G* may be removed from the bottom of the injector body.

Do not cut out the web in the barrel cap. It is put there to prevent the tubes dropping down should they become loose from vibration or expansion.

When reseating the automatic water valve disc, reseat also the bottom of the water valve stem to the inside valve disc guide seat. This assures tight closing of the valve, should the main boiler check leak. See that the port holes in the water valve disc guide are free and open.

Testing the Type "T" Injector. Start the injector in the regular way, close the overflow, and shut down on the water valve, thus putting steam pressure into the injector body. This will be a warning to be given through the telltale exhaust located in the cab. This indicates that the telltale would give warning if the injector fails to deliver water to the boiler when in road service. If the steam blow stops only when the heater valve is completely closed, the spring in the telltale is broken and should be repaired.

To operate the type "T" injector with closed overflow and with hot or cold water, proceed as follows:

(1) Turn on the steam and open the overflow and water valve in the usual manner.

(2) Pull the injector starting valve lever to full-open position.

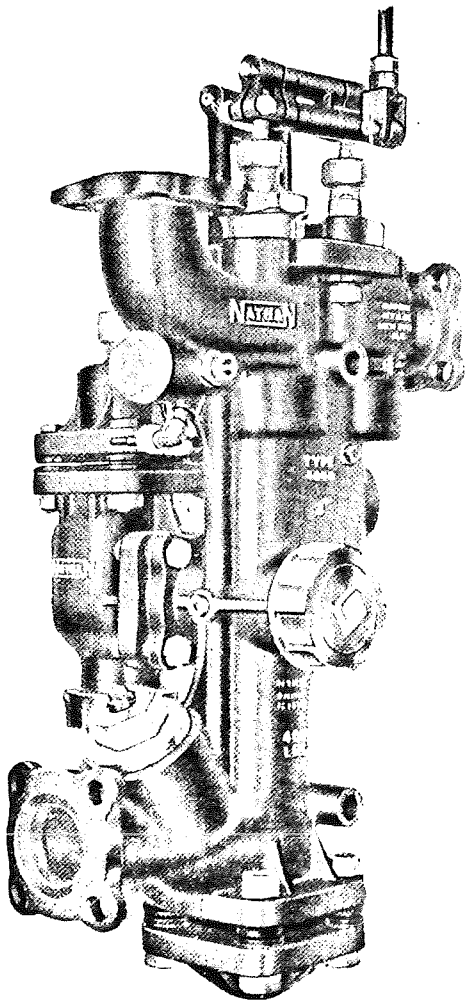


Fig. 31. Nathan Type 4000 Injector
Courtesy of the Nathan Manufacturing Co.

The injector will restart automatically even though the feed water supply be momentarily interrupted. The overflow is closed automatically when the injector goes into operation. It opens automatically before the injector "breaks" and prevents a steam blow back into the tank hose. When the water supply is reestablished, it restarts and closes.

Single-lever operation controls both water and steam. As the operating lever is moved to increase or decrease the injector capacity, the proper relation of steam and water is provided for automatically.

Operation of the "Type 4000" Injector. The initial movement of the cab operating lever primes the injector. It lifts the water, if the water level in the tank is below the water inlet in the injector. As the lever is pulled into upright position, the water valve and steam valve are both opened wide, and the injector operates at full capacity. Pulling the lever back of the wide-open position reduces the capacity to the desired amount. Water and steam are both regulated as the lever is hooked back. The injector is shut off by pushing the lever all the way forward.

A manual control is provided for closing the overflow valve by hand when it is desired to admit steam into the injector body and the water line to prevent freezing. The operating lever is then placed in the "priming" or heater position. In this position the water valve is opened and the steam circulates to heat the complete injector and the suction line. Even though an attempt were made to start the injector with the manual overflow closed, full pressure cannot be blown back into the tank hose as the water valve acts as a non-return check valve and closes the communication to the water intake line, permitting only a reduced amount of steam to enter around the bottom disc of the water valve. As soon as the pressure in the injector body rises to 20 pounds per sq. in., the overflow check valve opens automatically.

The type 4000 injector is both lifting and non-lifting and, therefore, it may be located above the lowest level of the tank when necessary.

The automatic restarting injector overflow arrangement, shown in Fig. 33, is applicable to other types of Nathan non-lifting injectors. When this device is used, the injector is operated, that is, started, stopped, and regulated in exactly the same manner as a conventional non-lifting injector. The operation of the arrangement is illustrated in Fig. 33. When the injector starting valve is opened, steam pressure enters through the top pipe connection to the top of the control piston *20,* holding the control piston down against its lower stop. The control piston keeps the relief valve *21* open, relieving any pressure from the top of the overflow operating piston *8* to the atmosphere. When the injector has started and after the delivery pressure acting on the lower side of the control piston *20,* communicated through the delivery pressure pipe, is higher than the steam pressure, the control piston will be lifted up and held against its top seat, permitting the relief valve to close and admitting the delivery pressure through passage *C* to the top of the overflow operating piston *8,* forcing it down and holding the overflow check *10* in closed position. The overflow operating piston in its closed position makes a tight seat at *E,* and the overflow check *10* is held to its seat by a precompressed spring *14* and by means of a spring follower *13.*

When for any reason the operation of the injector is interrupted, the delivery pressure acting on the

(3) When the injector starts to deliver water to the boiler, close the overflow valve.

As long as the injector delivers water to the boiler, there will be no steam discharge at the telltale nozzle.

The injector should then be shut off in the usual manner.

If the injector is to be re-started, proceed as described above.

To operate the type "T" injector with open overflow, proceed as follows:

(1) Turn on the steam with the main steam valve at the manifold.

(2) Turn the overflow valve to full-open position.

(3) Turn the water valve to full-open position.

(4) Pull injector starting valve lever to full-open position.

(5) After the injector is started, regulate the water supply with the water valve.

(6) When the injector is shut off, close the water valve and leave the overflow valve open.

To operate the heater, when the injector is not in operation and the feed line is in danger of freezing, proceed as follows:

(1) Open the water valve.

(2) Close the overflow valve.

(3) Open the heater valve *J*.

(4) If steam is discharged into the telltale drain funnel, valve *J* is too wide open and should be throttled down until the discharge stops.

Do not try to start the injector with the heater valve *J* open.

Do not try to start the injector with the overflow valve closed.

Do not put steam into the injector for heating to prevent freezing by opening the starting valve and regulating the main steam valve at the turret or manifold.

Overpressure developed by injector delivery should not exceed boiler pressure by more than 15 to 20 pounds. If this overpressure is higher than 20 pounds, look for delivery restrictions, such as scale in the delivery pipe, particularly at bends in the pipe. Also, examine the boiler check valve for valve action or lift and inspect the steam forcing tube of the injector.

Care should be taken to see that all pipe connections to the injector are in true alignment. This avoids strain to the injector body, which may cause bodies to break or distort and seriously affect the action of the injector tubes.

The operating rods on the steam water and overflow valves should be free from binding or strain from the injector to the cab. See that the operating rod travel is of correct length so that valves can be seated. Keep the overflow and water valves well packed and packing nuts drawn up.

Maintenance of the Standard "T" Lever Starting Valve. Refer to Fig. 24. In repairing or replacing the seat ring *J515-Y* in the starting valve, grind it in lightly after the seat ring has been screwed to its seat. If this ring is not properly applied, it is possible to distort it and it may leak even though it has been ground to a seat before application.

Nathan Injectors

A sectional view of the Simplex "type R" lifting injector is shown in Fig. 28. In the action of this injector, steam from the boiler is admitted to the lifting nozzle *22* by pulling the starting lever *4* slightly and without withdrawing the plug on the end of the steam spindle *11* from the steam nozzle *21*. This permits the steam to pass through the small openings around the steam nozzle and discharge into the overflow chamber, lifting the heater cock check *26* and issuing from the overflow.

When water appears at the overflow, the starting lever *4* is pulled back to the limit, thus opening the steam nozzle *21*, allowing a full supply of steam to enter the intermediate nozzle *23* and forcing the water through the delivery nozzle *25* into the boiler.

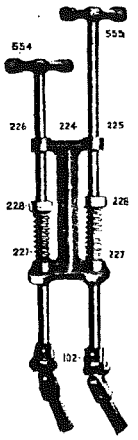
High steam pressures produce a vacuum in the overflow chamber which draws an additional supply of water into the nozzles through the inlet valve *19* and through the supply openings between the nozzles, thus increasing the capacity of the injector.

If the inlet valve should leak and prevent proper priming, the emergency valve *35* may be used to cut out the inlet valve, thus enabling continued operation of the injector until such time as the defective valve is repaired.

Figs. 29 and 30 show the Nathan non-lifting automatic high pressure injector type "HP."

The Nathan type 4000 injector is illustrated in Figs. 31 and 32. This new injector has several outstanding features, namely: The restarting closed overflow, positive one lever operation, automatic regulation of the steam supply in relation to the water supply, and a balanced starting valve applied directly in the injector body.

The automatic restarting overflow, shown in detail in Fig. 33, enables the injector to handle very hot tank water, and it permits a finer regulation of the water supply resulting in higher delivery temperatures.



Names of Parts

102	Universal Joint.
224	Stand
225	Rod for Waste
226	Rod for Water
227	Spring
228	Loose Collar
554	Water Hand-Wheel
555	Waste Hand-Wheel
556	Set Screw in 228

Fig. 8. Cab Stand for Non-Lifting Injector

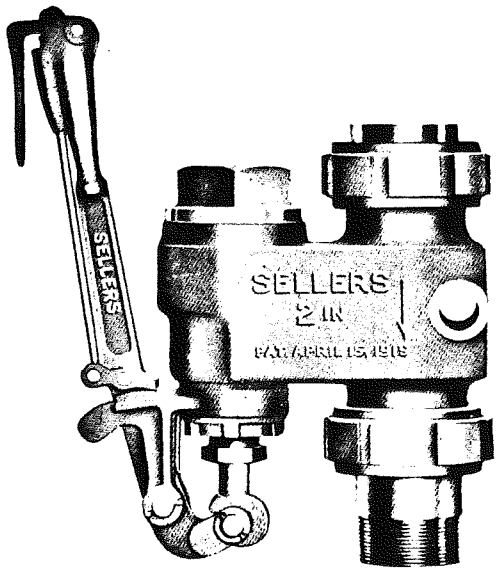


Fig. 9. Sellers Balanced Lever Starting Valve for Non-Lifting Injectors

mitted to escape without danger of bursting the tank hose.

When the starting valve is placed outside of the cab, an extension rod is used with a toothed handle sliding in a plate bolted to the boiler or cab frame as shown in Figs. 11 and 12. When the hand-pull is drawn back, one quarter rotation of the handle engages the end slot with the side of the bracket, holding the valve wide open. By turning to the right or left to engage with the intermediate slots, the amount of steam used for heating the feed water may be regulated.

When the starting valve is placed outside of the cab, the lever *604*, Fig. 11, must be at full open stroke when notch *A* of pull-handle *590* locks with the boiler bracket *591*.

The valve body may be placed as shown in the cuts, or it may be rotated flat against the boiler; the quadrant may be swiveled, so that the lever may be in convenient position for the engineer, or for the attachment of the extension rod and handle.

The Injector Indicator. The Sellers injector ndicator for non-lifting injectors, Fig. 13, notifies the enginemen when the injector wastes, "flies off," blows back, or is at minimum capacity.

The indicator consists of a cylinder containing a loosely fitting piston *661* with a projecting plug, visible when the injector is not feeding. It is connected to the overflow chamber of the injector body by a ½" copper pipe. When the injector is feeding properly, there is always a light vacuum in the overflow chamber and this lifts the piston of the indicator, and the plug is drawn up. The moment the injector starts to waste, the plug drops. The same is true if the injector "flies off" or blows back. The minimum capacity of the injector may be obtained by throttling until the plug moves rapidly up and down.

Installing the Non-Lifting Injector. The instructions previously given for the installation of lifting injectors will also apply to non-lifting injectors, except that the non-lifting injector should be applied vertically at the lowest water level of the tank.

Operating the K, N-L Injector. To start the injector, open the water valve and waste, and then pull the steam starting valve wide open. To stop the injector, close the steam starting valve and the water valve. To regulate the amount of water fed to the boiler, adjust the water valve only, the steam starting valve must remain wide open. To heat the tank water, close the waste valve, open the water valve and steam starting valve to the first notch. When feeding the boiler, the waste valve must be wide open, except when using very hot water.

The Sellers U. S. Standard injector has the same action as the class K, N-L.

Sellers Type "S" Injectors. The Sellers type "S" injector is illustrated in Fig. 14. This injector is a combined lifting and non-lifting type that may be located below the locomotive cab at any height above the rail. It utilizes all the water in the tank regardless of its level below that of the injector. Fig. 15 shows a typical application of the type "S" injector.